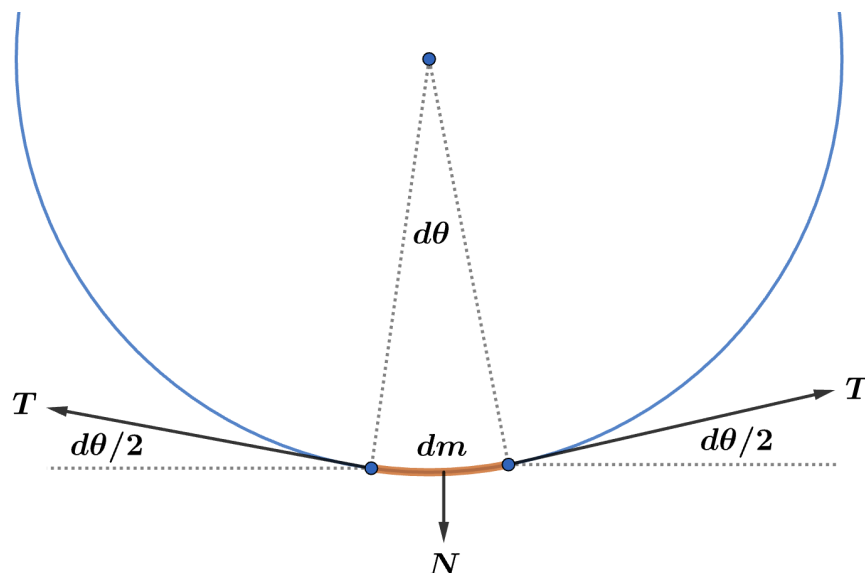


2022A F=ma Exam: Problem 25

Kevin S. Huang



Consider a small piece of the rubber band with mass dm . Balancing forces in the plane perpendicular to the axis of the cylinder, we have

$$N = 2T \sin\left(\frac{d\theta}{2}\right) = Td\theta$$

In the direction parallel to the axis, the friction force must balance the weight of this piece,

$$f = dm g \leq \mu N$$

Substituting for the normal force,

$$dm g \leq \mu T d\theta$$

The density of the rubber band is uniform,

$$\frac{dm}{d\theta} = \frac{m}{2\pi}$$

and our equation becomes

$$\frac{mg}{2\pi} \leq \mu T$$

The coefficient of friction satisfies

$$\mu \geq \frac{mg}{2\pi T}$$

so the answer is A.